

Multidefect Detection Tool for Large-Scale PV Plants: Segmentation and Classification

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Abstract—Unmanned aerial vehicles (UAVs) with high-resolution optical and infrared (IR) imaging have been introduced in recent years to perform inexpensive and fast inspections in operation and maintenance activities of solar power plants, reducing the labor needed, while lowering the on-site inspection time. Even though UAVs can acquire images extremely quickly, the analysis of those images is still a time-consuming procedure that should be performed by a trained professional. Therefore, a computer vision approach may be used to accelerate image analysis. In this work, a dataset of IR images was created from a 10-MW solar power plant and a comparative analysis between mask R-convolutional neural network (CNN) and U-Net was performed for two experiments. Concerning the defective module segmentation, the mask R-CNN algorithm achieved a mean average precision at intersection over union (IoU) = 0.50 of 0.96, using augmentation data. Regarding the segmentation and classification of failure type, the algorithm reached a value of 0.88 considering the same evaluation metric and data augmentation.

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When compared to the U-Net in terms of IoU, the mask R-CNN outperformed it with 0.87 and 0.83 for the first and second experiments, respectively.

Index Terms—Class of abnormality, convolutional neural network (CNN), failure mode, image classification, image segmentation, large-scale photovoltaic (PV) plant, thermographic inspection.

I. INTRODUCTION

PHOTOVOLTAIC (PV) solar energy's panorama has changed over the last few years. The high demand for renewable energy sources has led to higher investment in PV solar installations. As it is considered one of the cleanest sources of energy, it is estimated an exponential growth of PV solar energy in the near future, resulting in plants with installed capacity on the range of peak GW [1], [2]. However, at this scale, operation and maintenance (O&M) actions become unaffordable with extended human intervention because of labor costs; thus, alternative approaches are needed [3], [4].

The use of unnamed aerial vehicles (UAVs) has proven to be a reliable resource for infrared (IR) image acquisition of PV plants, allowing for optimization labor costs and duration and constituting a cheaper and quicker alternative for thermographic inspection through computer vision analysis techniques [5], [6], [7].

Different methods have been presented in the literature to detect and classify PV module failures through UAV's thermographic inspection, mainly module detection and thermal anomalies. Regarding PV module detection, which is identifying a module in an image, the majority of the work is based on the use of classical computer vision algorithms using binary thresholding as the most usual procedure in obtaining masks for their segmentation [8], [9], [10], [11], [12], [13]. Contour extraction followed by feature extraction followed by classification using the support vector machine represents an alternative [14]. Furthermore, other studies have employed morphological operations, Hough transform [15], and template matching [16]. However, the need to suppress the problems associated with the expense of algorithm optimization and the increase of new images have given rise to the implementation of deep learning techniques [17], [18]. On the other hand, regarding thermal anomalies, the detection of hot spots is the failure mode that has generated the most scientific interest. For this purpose, segmentation techniques by threshold and mask [15], [18], and template matching [16] have been used, as well as the identification of outliers [11] by extracting features, such as

TABLE I
DEEP LEARNING WORKS FOR MODULE DETECTION AND/OR FAULT
LOCALIZATION/CLASSIFICATION

Work	Methods		Anomaly classes
	Neural Networks	Task	
[1]	Mask R-CNN U-Net FPNet LinkNet	Defective Module Segmentation/ Localization	3
[14]	Mask R-CNN	Module Segmentation	-
[19]	Mask R-CNN + ResNet50	Module Segmentation + Fault Classification	10
This study	Mask R-CNN U-Net	Defective Module Segmentation + Fault Classification	5

mean and standard deviation in each module or comparing it to neighboring modules.

Deep learning approaches have also been increasingly used in image processing. Combined with UAV image acquisition, deep learning has overcome the limitations of classical image processing techniques [19], which typically require significant manual tweaking of parameters and result in poor generalization to unseen imagery [20], [21], [22], [23]. As an example, using a convolutional neural network (CNN) for imaging classification of defects and failure modes assumes a preponderant role, because of its ability to learn important properties for each class of defect without the need for human input. Moreover, other techniques, such as instance segmentation have also been applied for module detection and/or fault localization/classification [1], [14], [19], as summarized in Table I.

The study reported by Pierdicca et al. [1] employs several networks (mask R-CNN, U-Net, FPNet, and LinkNet) to locate/segment defective cells, categorizing all images as containing anomalous cells, multiple anomalous cells, and contiguous series of anomalous cells. The work presented by Vega Diaz et al. [14] relies on module segmentation using mask R-CNN to distinguish foreground (modules) from background. In Bommies et al. [19], the authors present the classification of all PV modules, after segmentation, into ten anomalies by utilizing two independent networks. Taking into account the major contributions of the mentioned works, in this letter, a deep learning-based approach for segmentation and classification of PV module failures for large-scale plants is developed, based on the preliminary work presented at the IEEE Photovoltaic Specialists 49th conference [24]. The main contributions of this research can be outlined as the use of segmentation of defective modules and their classification into five failure types with the introduction of class of abnormalities. For that, a comparative analysis of performance between two networks, mask R-CNN and U-Net, is performed.

TABLE II
THERMOGRAPHIC INSPECTION DETAILS

Parameter	Range
Module Emissivity	0.95
Air Temperature (°C)	35
Wind Velocity (km/h)	10–15
Cloud Coverage	Blue sky
Distance (m)	23–25
Camera	Thermal and Visual
Irradiation (W/m ²)	700–1000
Thermal Camera Resolution (pixels)	640 × 512

II. THERMOGRAPHIC INSPECTION

Solar cells with defects emit different IR radiation patterns, allowing the IR imaging to be used to scrutinize PV failure modes. For thermographic inspection of PV modules, IEC TS 62446-3: 2017 defines the required external conditions [25].

A. Environment Conditions

The environmental conditions must be taken into consideration when conducting local IR measurements. Therefore, the following conditions should be met.

- 1) Minimum level of irradiation: 600 W/m².
- 2) Maximum wind speed: 28 km/h.
- 3) Maximum cloud coverage: 2 okta.
- 4) Low or no soiling.

B. Classes of Abnormalities (CoA)

Depending on the temperature difference between the failure and healthy modules, there are several different categories of abnormalities.

Accordingly, three classes of abnormality were considered for the methodology presented in this work as follows.

- 1) CoA 1: No abnormality, no action required.
- 2) CoA 2: Thermal abnormality, rectification is advised.
- 3) CoA 3: Safety-relevant thermal abnormality, operational interruption is required.

III. METHODOLOGY

A. Dataset

A thermographic inspection was conducted using the DJI S1000 drone, equipped with a Workswell WIRIS second generation camera containing both visible (VIS) and thermal lens. Each image was acquired at a constant speed of 3 m/s. The angle between the modules and the camera was kept higher than 30°, in order to minimize the optical reflection. This process was carried out in accordance with the environmental conditions previously enumerated.

The inspected PV plant is in Beja, Portugal, and it consists of 42 048 modules (CSUN240-60P and GOOSUN GS230) installed at an angle of 30°. It was installed between 2010 and 2012 and has a peak power capacity close to 10 MW. Table II summarizes the environmental and technical conditions at the

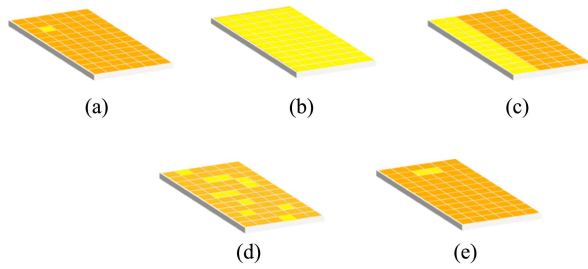


Fig. 1. Classification of modules defects in thermography analysis. (a) Hot spot. (b) Module in open circuit. (c) Substring in open circuit. (d) Module in short circuit. (e) Heated module junction box.

TABLE III
IDENTIFICATION AND QUANTIFICATION OF DEFECTS

Type of defect	Quantity of detected defects
Substring in open circuit	172
Single cell with difference in temperature	318
Modules in open circuit	469
Heat module junction	10
Module in short circuit	26

time of the thermographic and visual inspection that resulted in the definition of the database.

A total of 1368 images consisting of several strings were examined and labeled by a human operator. The founded defects are gathered in five categories, as illustrated in Fig. 1. A relabeling procedure was then performed by other three operators for higher consistency of the initial label.

Considering that each image consists of several modules, each may contain more than one anomaly. Also as these images are from an operating plant, the dataset is limited by the presence of reflections, shadows, and bird drops, which may result in assessment errors or misinterpretations during labeling. Therefore, as part of the training and evaluation process of the algorithm, several images were discarded from the original dataset in order to minimize error propagation during the image analysis. A total of 46 images have been removed due to reflection issues, resulting in 155 images for consideration involving five types of defects, as described in Table III.

B. Data Augmentation

There are a few challenges associated with the defined dataset, such as its small size and the large imbalance between the types of defects. In this sense, data augmentation was introduced to enlarge the dataset size and to allow our model to generalize from multiple future perspectives, i.e., to simulate the acquisition of images from various viewpoints from which an image may be acquired. A number of transformations have been performed upon the image during the augmentation process, including affine transformations (scaling, translation, rotation, shear), changes in luminosity, and the addition of noise. Through this approach, new and different images were obtained from the original ones.

The augmentation methods on-the-fly and offline were used in U-Net and mask R-CNN, respectively.

C. Deep Learning Approaches

It has been demonstrated in a number of studies that instance segmentation, namely, mask R-CNN [26], is effective for the segmentation of PV modules and/or the detection of failures. Developed on top of the faster R-CNN [27], a known network used in object localization, the mask R-CNN provides an additional branch for mask prediction in addition to the existing branch for bounding box prediction. In this work, mask R-CNN was trained with a ResNet-50 as backbone using Detectron2, the latest Python library for object detection. The hyper parameters, such as batch size, number of region of interest heads, and region proposal network ratio were then adjusted to optimize the model. Notwithstanding, another approach can be used to the task of semantic segmentation on top of mask R-CNN, a neural network architecture known as U-Net. Based on biomedical imaging segmentation [2], this architecture consists of an encoder network followed by a decoder network. In the encoder section, important features are extracted from the images, whereas in the decoder section, the network is concatenated with the previous layers in order to merge the feature and the localization information.

Two experiments were conducted using the aforementioned networks. In the first experiment, the localization of defective modules and the module segmentation tasks using mask R-CNN were combined to evaluate the effectiveness of the mask R-CNN and U-Net in detecting defective modules. The second experiment leverages the first one with the addition of classification on five types of failure modes and corresponding CoA. A schematic representation of the entire process is shown in Fig. 2.

For the purpose of evaluating the model accuracy, the dataset was divided into two categories with 90% of the dataset being used for training, and 10% being used for validation. As a result of the dataset's inherent imbalance, a stratified split was applied to ensure that the support ratios on the training and validation sets are the same. This analysis was performed using a workstation with an Intel(R) Xeon(R) Gold 6140 CPU 2.30 Ghz processor, 32 GB RAM, and NVIDIA GeForce RTX3090 GPU.

C. Evaluation Metrics

The proposed methodology was evaluated common metrics for image segmentation and object detection, the intersection over union (IoU) and the mean Average Precision (mAP).

An IoU threshold of 0.50, expressed by (1), was defined to measure the mAP of the proposed approach

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}. \quad (1)$$

Based on this definition ($\text{IoU} = 0.50$), we calculated the Precision and Recall of the model using (2) and (3), respectively, by determining the amount of TP (True Positives), FP (False Positives), and FN (False Negatives)

$$\text{Precision } (P) = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall } (R) = \frac{TP}{TP + FN}. \quad (3)$$

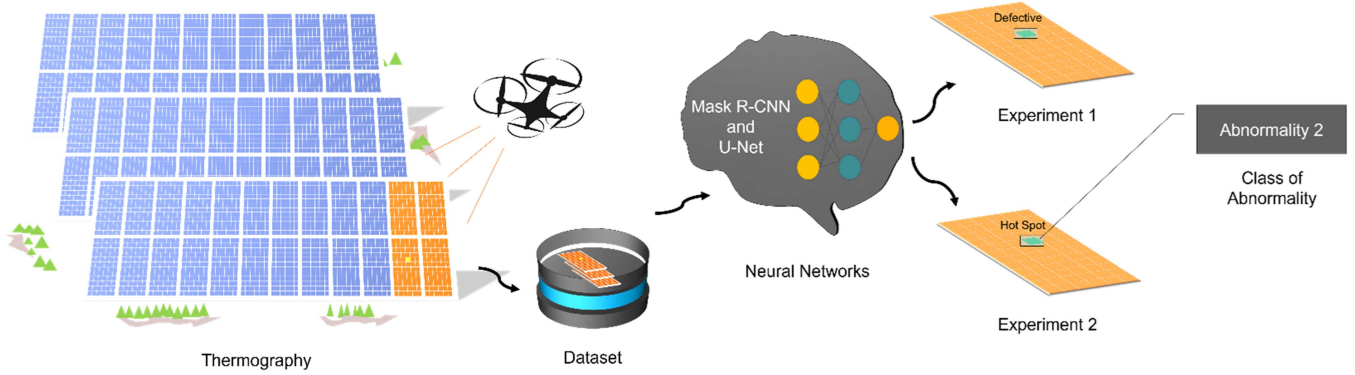


Fig. 2. Schematic representation of the proposed methodology.

TABLE IV
CONTRIBUTION OF AUGMENTATION TO THE RESULTS OF MASK R-CNN

Segmentation mode	mAP at IoU = 0.50	
	w/o augmentation	w/ augmentation
Experiment 1: Segmentation of PV module defective	0.93	0.96
Experiment 2: Segmentation of Failure Type	0.73	0.88

A precision–recall (PR) curve is generated for each object class and the area under that curve represents the Average Precision (AP) of the model. The mAP is then obtained through the mean of the AP for all object classes.

IV. RESULTS AND DISCUSSION

The pretrained mask R-CNN was first fine-tuned for both experiments and evaluated using the mAP and IoU, as described in Table IV.

Based on the results of experiment 1 and by using data augmentation an improvement of 0.93–0.96 is achieved. Similarly, experiment 2 confirms the same findings: with the use of data augmentation, an increase of 0.73–0.88 is achieved. Such improvement demonstrates the data dependency of the failure identification task. Comparing the two experiments, the mask R-CNN achieved a mAP of 0.96 and 0.88 at an IoU threshold of 0.5, for the first and second experiments, respectively. These results demonstrate that experiment 2 task is more challenging and additional data may be required.

Table V provides a comparison of mask R-CNN and U-Net with regard to segmentation for both experiments. It is evident that U-Net has a more challenging task to accomplish, since it is purely a segmentation network. Mask R-CNN presents the same performance with and without data augmentation in the first experiment, achieving 0.87 in both cases. When compared with the U-Net, the performance decreases to 0.82.

In the second experiment, the use of data augmentation contributed for a better performance in mask R-CNN, increasing

TABLE V
IOU VALUES FOR DIFFERENT NETWORKS

Segmentation mode	IoU		
	Mask R-CNN		U-Net
	w/o augmentation	w/ augmentation	w/ augmentation
Experiment 1: Segmentation of PV module defective	0.87	0.87	0.82
Experiment 2: Segmentation of Failure Type	0.77	0.83	0.75

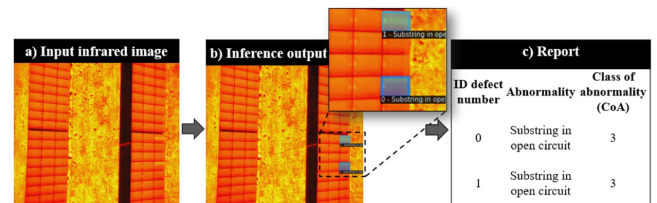


Fig. 3. Workflow process of the proposed methodology. (a) Input IR image. (b) Inference output. (c) Description of defect type and corresponding CoA.

from 0.77 to 0.83. A similar loss performance behavior was observed with the U-Net for experiment 2, resulting in a value of 0.75. Based on these results, it is concluded that the mask R-CNN performs better than the U-Net, even without data augmentation.

Actually, mask R-CNN consistently outperformed U-Net for both experiments, demonstrating the importance of data augmentation for this application, especially when dealing with classification into failure types.

The mask R-CNN model was therefore, selected for the identification of the class of abnormality. Fig. 3 describes the entire process for a sampled image. The IR image of a section of the PV plant is uploaded, analyzed by the pipeline to determine the defects and then displayed in a new image. As a result, the output data can be used to identify the failure type and the class of abnormality.

Generally, the image was submitted for inference which results in instance segmentation of the failure types. Afterward,

on the basis of the failure type, the CoA was extracted taking into account the temperature difference between the healthy and the failure modules.

V. CONCLUSION

In this work, a novel approach for the segmentation and classification of PV modules with the identification of classes of abnormalities was proposed. Moreover, a new dataset was created.

Two experiments were conducted to evaluate the used approach. The first experiment considered the identification and segmentation of defective modules, whereas the second one was related to the classification of failure types. The latter had the main advantage of identifying separate defect types on PV modules and consequently its CoA characterization. In addition, the metrics evaluated have deteriorated in the second experiment, demonstrating the difficulty of identifying separate defects. According to the results obtained from the comparative analysis of mask R-CNN and U-Net, the first method obtained better results than the second.

The real-life collected images still have limitations that make them hard to evaluate, such as reflections and shadows. As a result, these limitations negatively impact a trained human operator to a small extent, but if not taken into consideration, could have a more significant impact on automatic fault detection. Thus, the relabeling of the dataset by experts contributes greatly to the improvement of the final results since it improves the data labeling consistency, while at the same time lowering the total amount of human labor. Despite trained drone operators following a standardized but complex image acquisition procedure, for the current set of algorithms, there is a necessity of performing more reliable image acquisitions.

The approach developed in this work offers the major advantage of avoiding the need to segment the image between foreground and background, as well as avoiding segmentation of all PV modules. In addition, the introduction of abnormalities allows the assessment of the danger levels involved on the O&M procedures.

It is intended for future work to consolidate this approach by combining electrical and electroluminescence data, contributing to a comprehensive assessment of the entire power plant.

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